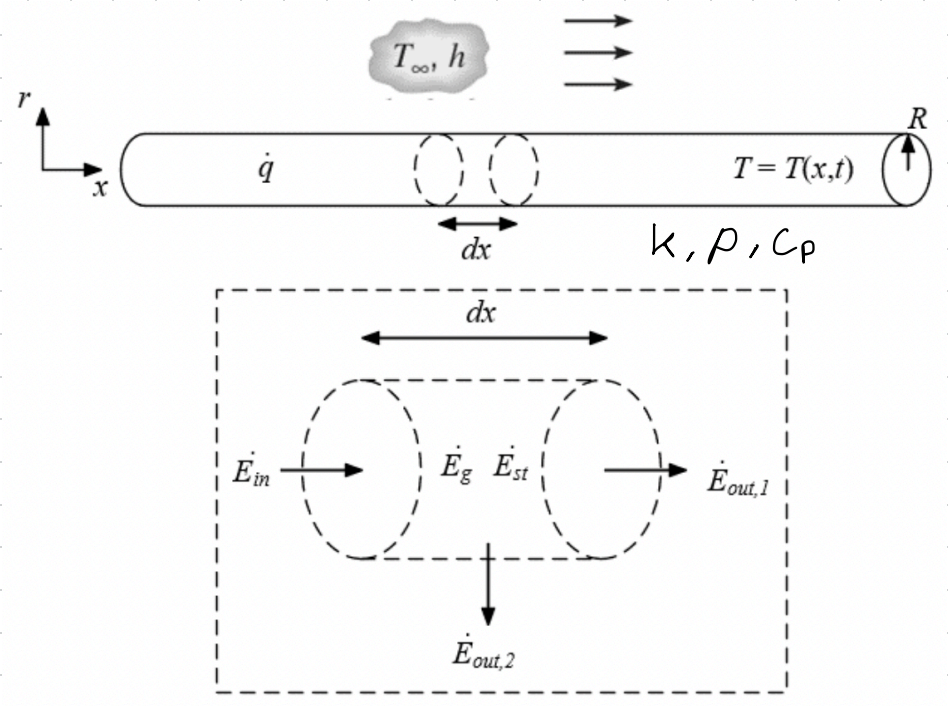
1) **Known:** cylinder radius, ambient temp, vol heat gen rate, h.t. coeff

Find: a) energy balance eqn for CV as a fctn of T
b) gov diff eq that describes temp profile
c) BCs, ICs for gov diff eq

Schematic:



Assumptions: very long and small radius rod, temp of rod radially uniform, neglect thermal radiation, thermal conductivity and density and heat capacity cst, temp of rod is a fctn of x & t.

Analysis: a) $\dot{E}_{in} = \dot{E}_{out,1} + \dot{E}_{out,2} - \dot{E}_g + \dot{E}_{st}$

$$q_x = q_{x+dx} + q_{conv} - \dot{q} + \rho c_p \pi R^2 dx \frac{\partial T}{\partial t}$$

$$q_x = -k(\pi R^2)\frac{\partial T}{\partial x}$$

$$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx$$

$$q_{conv} = h(2\pi R)\,dx\,(T - T_\infty)$$

$$\cancel{q_x} = \cancel{q_x} + \frac{\partial q_x}{\partial x}\cancel{dx} + h(2\pi R)\cancel{dx}\,(T - T_\infty) - \dot{q}\pi R^2 \cancel{dx} + \rho c_p \pi R^2 \cancel{dx}\frac{\partial T}{\partial t}$$

$$\boxed{kR\frac{\partial^2 T}{\partial x^2} = 2h(T(x,t) - T_\infty) - \dot{q}R + \rho c_p R\frac{\partial T}{dt}}$$

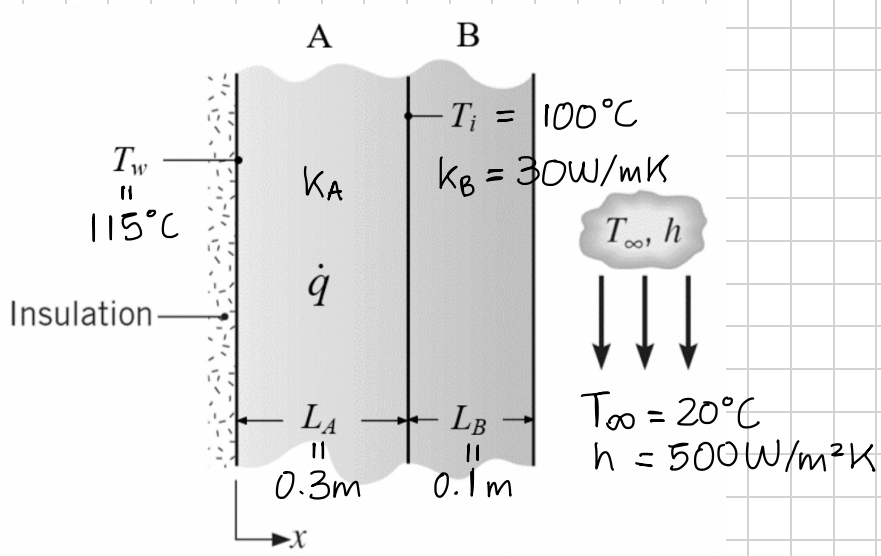
b) $$\boxed{\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}\left(T(x,t) - T_\infty\right) + \dot{q}}$$

c) $$\boxed{\begin{aligned} &T(0,t) = T_1 \\ &-k \left.\frac{dT}{dx}\right|_{x=L} = h\left(T(L,t) - T_\infty\right) \\ &T(x,0) = T_i \end{aligned}}$$

2) **Known:** dimensions of Wall A & B, vol heat gen, thermal conductivity of A & B, coolant temp, convective h.t. coeff, T_w & T_i from sensor

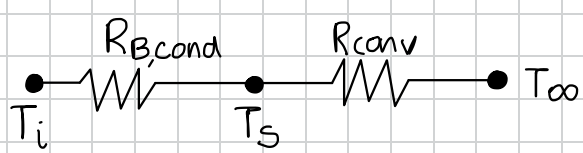
Find: a) right surface temp T_s of Wall B
b) uniform volumetric heat gen, $\dot{q}$ @ S.S.
c) k_A
d) plot temp profile across Wall A & B with important features.

Schematic:



Assumptions: 1D heat conduction, uniform $\dot{q}$, left surface of Wall A thermally insulated, no heat gen in Wall B, A & B in contact with no R_{th} @ interface

a) $q = \dfrac{T_i - T_\infty}{R_{tot}}$, $R_{tot} = R_{B,cond} + R_{conv}$



$$T_i - T_s = q\, R_{B,cond}$$

$$T_i - T_s = \frac{(T_i - T_\infty) R_{B,cond}}{R_{B,cond} + R_{conv}} = \frac{(T_i - T_\infty)\, L_B/k_B}{L_B/k_B + 1/h}$$

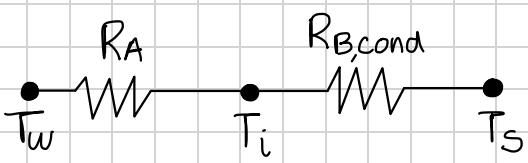
$$T_s = T_i - \frac{(T_i - T_\infty)\, L_B/k_B}{L_B/k_B + 1/h} = \boxed{50°C}$$

b) $q = \dot{q} L_A$

$$(T_i - T_\infty)/(L_B/k_B + 1/h) = \dot{q} L_A$$

$$\boxed{\dot{q} = 5 \times 10^4 \ W/m^3}$$

c)



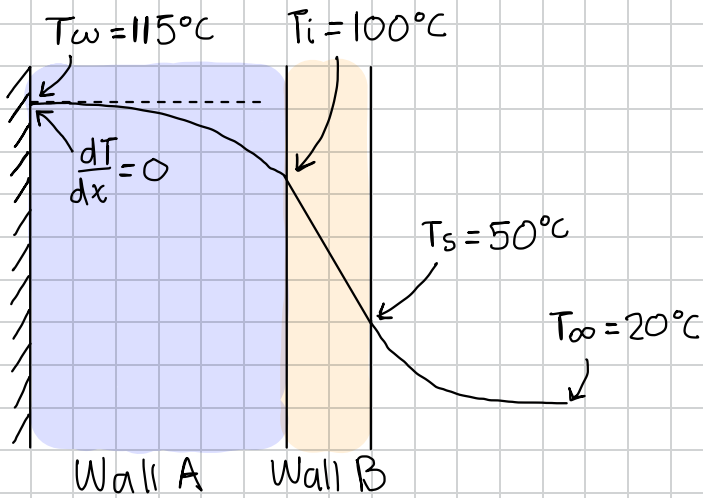
$$T_w - T_i = \frac{(T_w - T_s)\, R_A}{R_A + R_{B,cond}}$$

$$T_w - T_i = \frac{(T_w - T_s)\, L_A/2k_A}{L_A/2k_A + L_B/k_B}$$

plug & chug

$$\boxed{k_A = 150\ W/mK}$$

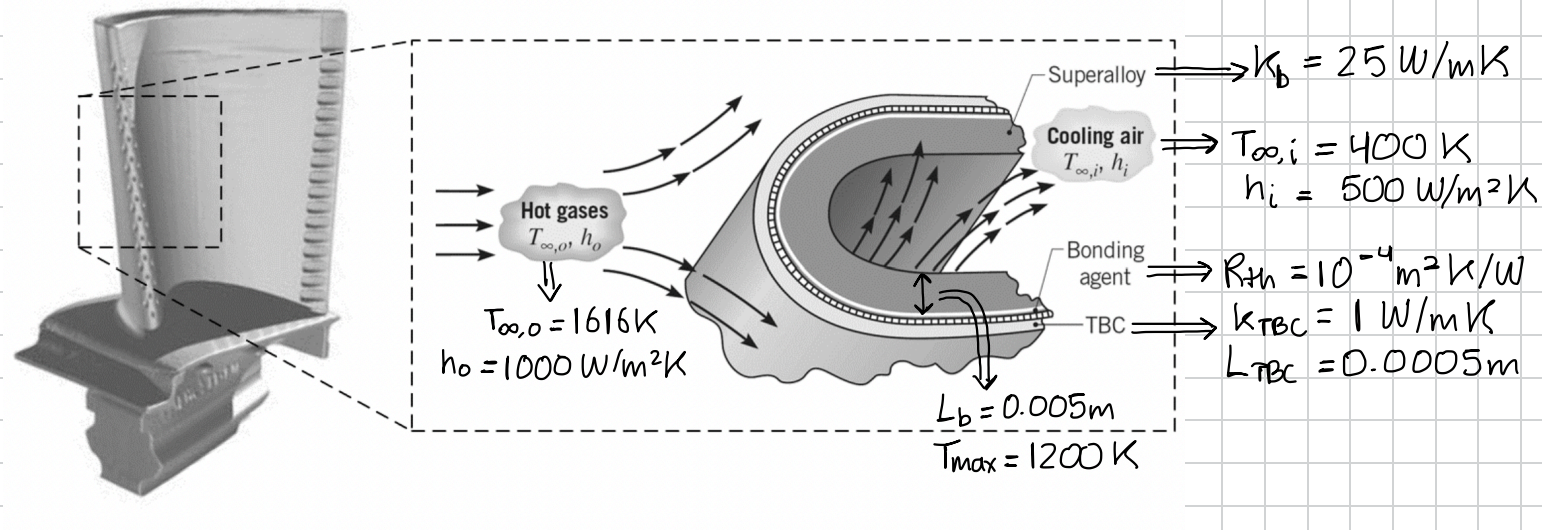
d)



3) **Known:** blade & TBC thermal conductivity, bonding agent R_{th}, external gas & internal air temp & h.t. coeff, blade thickness, max allowable blade temp.

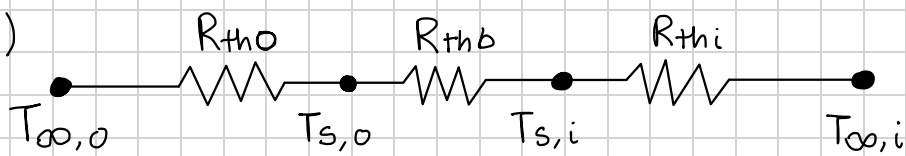
Find: a) w/ no TBC, draw R_{th} network & label, T_s, can $T_s < T_{max}$?
b) w/ TBC, same as a)

Schematic:



Assumptions: negligible thickness of bonding agent, S.S., thermal radiation negligible, turbine blade approx as plane wall, 1D heat conduction thru TBC & superalloy

Analysis: a)



$R_{tho} = 1/h_o = 0.001\ m^2K/W$
$R_{thb} = L_b/k_b = 0.0002\ m^2K/W$
$R_{thi} = 1/h_i = 0.002\ m^2K/W$

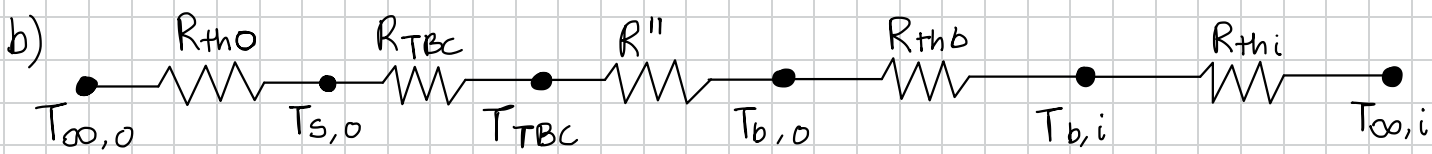
$R_{th} = 0.0032\ m^2K/W$

$$q = \frac{T_{\infty,o} - T_{\infty i}}{R_{th}} = 3.8 \times 10^5\ W/m^2$$

$$T_{s,o} = T_{\infty,o} - q R_{th} = \boxed{1236\ K}$$

$1236K > 1200\ K \Rightarrow$ **No, blade cannot be maintained below T_{max}**

b)



$R_{thO} = 1/h_o = 0.001\ m^2K/W$
$R_{thb} = L_b/k_b = 0.0002\ m^2K/W$
$R_{thi} = 1/h_i = 0.002\ m^2K/W$
$R_{TBC} = L_{TBC}/k_{TBC} = 0.0005\ m^2K/W$
$R'' = 10^{-4}\ m^2K/W$

$R_{th} = 0.0038\ m^2K/W$

$$q = \frac{T_{\infty,O} - T_{\infty i}}{R_{th}} = 3.2 \times 10^5\ W/m^2$$

$T_{s,o} = T_{\infty,O} - q R_{thO} = 1296\ K$

$T_{TBC} = T_{s,o} - q R_{TBC} = 1136\ K$

$T_{b,o} = T_{TBC} - q R'' = \boxed{1104\ K}$

$1104\ K < 1200\ K \Rightarrow$ Yes, blade can be maintained below T_{max}